

Bloch — Tokenomics V4 (Genesis-4 relaunch with carryover)

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BLOCH-TOKENOMICS-V4

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DRAFT — founder decision recorded, parameters not frozen

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tokenomics_v2.rs (21 B nominal, uncapped tail) and ADR-035 emission V3

Relates to:

BLOCH-POS-SHA3-LATTICE-MIGRATION.md (§4 distribution gates)

1. Decision

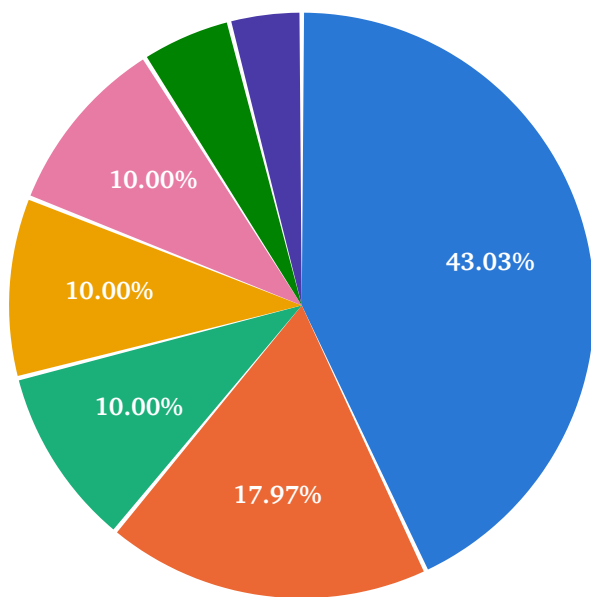
Relaunch from a fresh genesis with a **fixed 21,000,000,000 BLCH** supply — the V2 nominal, after a draft at 100 billion.

The whole carryover comes across as one balance set, with no founder line. Those coins were mined, on the same chain, under the same rules as everyone else's — so they are carried the same way, as ordinary liquid balance. The founder additionally receives a new 17% grant under a 10-year cliff and 40-year linear vest — the V2 premine schedule, at 10% rather than the V2 17%.

Returning to 21 billion removes two hazards the 100-billion draft created, at no cost: the supply is **11.38% of u64::MAX** rather than the earlier draft's 54.21%, so the sum of two large balances is nowhere near the wrap point; and it **fits in the signed int64** the Go SDK uses for `Satoshis ( sdk/go/models.go:16 )`, which 100 billion overflowed by 8%.

Destination	BLCH	Share	Unlock
Carryover — the whole ledger	3,773,884,800	17.97%	liquid at genesis
Founder — new grant	2,100,000,000	10.00%	10-year cliff, then 40-year linear
VC / crypto hedge funds	2,100,000,000	10.00%	12-month cliff, then 24-month linear
Development team	2,100,000,000	10.00%	18-month cliff, then 36-month linear
Marketing	840,000,000	4.00%	25% at genesis, remainder linear over 24 months
Liquidity	1,050,000,000	5.00%	100% liquid at genesis
Validators	9,036,115,200	43.03%	emitted over 40 years
Total	21,000,000,000	100.00%	

Founder total: 26.89% — the carried-over balance plus the new grant. The grant was cut from 17% to 10% on 2026-08-11 and the 1,470,000,000 difference went to validators, the only reallocation so far that moved supply away from an insider bucket.



Distribution of the 21,000,000,000 BLCH supply. Slices carry a direct label only where one fits; every figure is in the legend, which is also the table view — three buckets sit at exactly 10% — a pie cannot show that as well as the number does, which is why the legend carries both.

Validator emission runs for 40 years and is supplemented by transaction fees. **After the 21 B is fully issued, validators are paid 100% from fees.**

2. What the carryover actually contains — measured live

Not estimated. A read-only UTXO snapshot was taken on node4 at **height 43,172** with `bloch-snapshot-utxo`, then aggregated by address:

UTXOs	448,337
Addresses	15
Total carried over	3,773,884,800 BLCH
Snapshot root (SHAKE-256)	280d604b32525f03...
Carryover digest (SHAKE-256)	92918209a106f297...

All of it crosses. There is no founder line and no exclusion list: those coins were mined, on the same chain, under the same rules as everyone else's.

The distribution inside the set is what it is, and is worth stating rather than leaving to be discovered:

	BLCH	Share of carryover
Largest single address	3,546,175,400	93.96%

	BLCH	Share of carryover
The other 14	227,709,400	6.04%

An earlier draft of this section reported 413,743 UTXOs across five addresses, from the Genesis-1 file rather than the live chain. Ten more addresses have accumulated balances by mining since Genesis-3, and the totals moved with them. Anything quoted from the older figures is stale.

3. The cap, retired

Earlier drafts capped carried-over balances at 300 M and excluded the founder from the carryover, and §3 tracked how close that cap was to binding. Both went with the single-set decision (§1): the cap existed to bound what legacy holders received *while the founder was excluded*, so with nobody excluded it would either bind on everyone — scaling every balance down by roughly 92% — or on no one.

Three things follow, and the third is the one worth keeping:

1. No pro-rata scale-down runs. Every holder keeps 100% of their position.
2. The measurement that mattered — whether third-party mining would push the non-founder total past 300 M before the terminal height — no longer decides anything. It was going to be close: measured 227.7 M at height 43,172 against a 300 M ceiling, growing about 4 M/day.
3. **The exclusion list is gone, and with it an unaudited power.** An earlier draft flagged that whoever writes the list decides who counts as founder, that nothing in the protocol checks it, and that it should therefore be published for challenge before the height passed. There is now no list to write. That concern was not mitigated — it ceased to exist, which is the only one of this document's risks that got resolved rather than traded.

3.1 Snapshot height — decided: 80,000

Measured at height **40,424**: non-founder holdings $\approx$ **236.8 M BLCH**, 79% of the cap, growing $\approx$ **3.97 M/day**. On the central estimate the cap binds around height **86,300**, roughly 16 days out.

The height must land before the cap binds, and that is the whole argument. Below the cap every holder keeps 100% of their position and no scale-down runs at all. Above it, the 300 M becomes a fixed pot shared pro-rata, and every coin mined after that point dilutes everyone — individually rational to chase, collectively worthless, and a pure race.

Height	Days out	Preserved at r=12%	r=16%	r=20%	r=25%
60,000	6.8	100%	100%	100%	98%
70,000	10.3	100%	100%	100%	91%
80,000	13.7	100%	100%	95%	86%
86,000	15.8	100%	100%	92%	83%
100,000	20.7	100%	94%	86%	77%

`r` is the share of emission going to third-party miners. The central estimate, 16.4%, is derived from a single prior measurement rather than a fresh audit, so the table spans the plausible range.

80,000 gives about two weeks of notice — the recognisable norm for a snapshot — on a round, legible number, and preserves every holder in full unless third-party mining is running well above the estimate.

Longer notice is actively worse here, which is counterintuitive. Holders need do nothing: balances are captured on-chain automatically, there is no claim and no migration. So the notice period buys transparency, not time to act — and the one action it does enable is accumulating more coins before the cut, which dilutes everybody. The usual "give people plenty of warning" instinct inverts.

Before announcing, measure the non-founder total against live balances rather than trusting the 16.4% estimate. If the real rate is above ~20%, drop to 70,000.

3.2 The chain halts at the snapshot

Height 80,000 is a terminal height, not just a measurement point. The current chain stops producing blocks there; Genesis-4 launches from the snapshot roughly six months later, after code review.

This resolves the dead-period problem rather than mitigating it. Had the chain kept running, the 166 days between snapshot and launch would have meant 4,016 M BLCH mined by people receiving nothing — and, worse, a rational miner switches off the day after the snapshot, leaving the network without hashrate during exactly the six months it still has users, wallets and an explorer pointed at it. Halting removes both: nobody mines coins with no future, and nobody buys into a chain with no future.

Two things must exist **before** height 80,000, which is about two weeks away.

3.2.1 The halt has to be a consensus rule

A chain does not stop because it was announced. Blocks above the terminal height must be **invalid**, shipped in a release and running on the fleet before the height arrives. Otherwise miners simply continue and the "halt" is a fork nobody agreed to.

This is a flag day in reverse and it inherits every flag-day hazard this project has already lived through: the release must actually be the binary the fleet runs. Anyone who does not upgrade will keep mining past 80,000 on a fork; that is tolerable, but only if the canonical snapshot is fixed at 80,000 and said so publicly.

3.2.2 After the halt, the chain's own history stops being evidence

This is the non-obvious one. PoW security is bought with ongoing hashrate. The moment mining stops, the cost of rewriting history from below height 80,000 collapses toward zero — anyone with modest hashrate can produce an alternative chain ending at 80,000 with different balances, and after a few months of no honest mining, it may carry more accumulated work than the real one.

Therefore the **signed snapshot artifact is canonical, not the chain**. At the halt, produce the balance set, hash it, sign it, and publish the digest widely enough that it cannot be quietly replaced — the same pattern already used for `carryover.tsv.gz` and its `.sha256`. Genesis-4 must be built from that artifact, and the artifact's digest should appear in the Genesis-4 genesis block itself. A chain nobody is defending is not a record.

One trust point, named. The taint list — which addresses count as founder — is set by the founder. Nothing in the protocol stops founder-controlled coins from being presented as third-party holdings and capturing part

of the 300 M. The list should be published with the announcement so it can be argued with before the height passes, not after.

3.3 Genesis bootstrap — who produces block 1

The adversarial review found a gap the halt decision opened and nothing closed. The original migration design seeded the validator set during a **hybrid PoW phase**: miners kept producing while deposits accumulated, so PoS activated onto an existing, funded validator set. Halting Genesis-3 and launching Genesis-4 from a snapshot deleted that phase. Genesis-4 has no PoW, and deposits are transactions — which need blocks, which need validators.

The chain cannot start. This is not a subtle failure: without an initial validator set there is no proposer for slot 0.

The fix: a genesis validator set

The genesis block carries an initial validator set the same way it carries the allocations — as consensus data, active from slot 0, with no deposit transaction required because there is no chain yet to carry one. Every later validator joins through the ordinary deposit path.

Three things this must satisfy, and one it must not pretend to.

It must be funded from a named bucket. Liquid supply at genesis is 5,033,884,800 BLCH: the whole carryover (17.97%), liquidity (5%) and the marketing TGE tranche (1%). VC, team and the founder's new grant are entirely cliffed.

That leaves two candidate sources, and the choice is not neutral. Genesis validators can stake from the **Foundation's** liquid holdings, or from the **carryover** — which in practice means the largest carried-over address, since it holds 93.96% of the set. Funding them from the carryover would put the founder's balance directly into consensus at slot 0, which is the §4.2 problem in its most concentrated form. Funding them from the Foundation keeps that door shut and is the assumption the rest of this section makes. Either way it has to be said plainly rather than shown as an unexplained line in the genesis file.

It must be large enough to be meaningful and small enough to be honest. The partition in §6.5.3 cuts the active set into 32 committees, so a set below 32 leaves empty committees and slots with no attestors. A floor of **64 genesis validators** gives every slot at least two.

It must be replaceable. Genesis validators exit through the ordinary path as independent stake arrives. Nothing in consensus privileges them after genesis.

3.3.1 Who operates them, and the one-year rule

The founder funds and operates the genesis cohort (founder decision, 2026-08-11). There was no alternative that did not amount to the same thing: liquidity and marketing are the Foundation's, and the Foundation is the founder until the board exists.

The commitment attached to it is that the start does not persist: **within one year the founder's consensus weight falls below one third**, and that is written as a consensus rule rather than a promise (`crates/bloch-pos-committee/src/genesis_cohort.rs`).

Why one third and not something rounder. It is the share that can stall a two-thirds quorum. Above it the founder can freeze the chain unilaterally; below it the founder cannot. The rule says, in one number: *after year one, the founder cannot halt Bloch alone*. Note what it does not say — one third is the **liveness** threshold, not the safety one, which is two thirds. Falling under a third stops the founder stalling finality; it was never what stopped anyone finalising a bad state.

How it is enforced. The cohort is a fixed set published in the genesis block, which can only shrink — members leave by exiting and nothing can be added. Their combined weight is capped, and the cap tapers linearly from the whole set at genesis to one third at one year, then holds:

Month	Cohort cap
0	100.0%
3	83.3%
6	66.7%
9	50.0%
12	33.3%

The taper is deliberate. A step from 100% to 33% on one epoch boundary would eject two thirds of the chain's consensus weight in sixteen minutes — the chain would stop the moment the rule bit. Tapering gives independent stake a year of steadily rising room instead of a single deadline.

Where enforcement stops, plainly. The cap binds the *cohort*. Nothing prevents funding new validators after genesis under addresses that are not in it, and no on-chain rule can see beneficial ownership behind an address. Past the cohort, one third is a commitment verified externally — by the Foundation's reporting rule and by whoever is watching. The enforceable part is real and bounded; claiming more would be claiming that consensus can see through an address, and it cannot.

What it must not pretend

Genesis-validator stake does not count toward gates G1–G4. A Foundation- funded set spread across 64 records reads as 64 independent participants to `top_share_bps` and `nakamoto_coefficient`, which measure the operator view — they cannot see one beneficial owner behind many records, and no on-chain metric can. The same reporting rule already written for the delegation program (`BLOCH-ENTITY-STRUCTURE.md` §5.1) applies here and for the same reason: the gates are measured on stake whose beneficial owner is not the Foundation, the founder, or Postern Labs.

So the honest statement of the launch is: **the chain starts centralised, by construction, and the gates measure the distance from there**. A hybrid PoW phase would have bought a genuinely independent initial set; the halt bought a clean break instead. That was a defensible trade, but it was made without noticing this was part of the price.

4. What this does to existing holders

Stated plainly, because it affects parties who are not in the room:

	Today	After V4
Coins carried over	3,475,441,200 (G-1 file)	3,773,884,800 (measured live)
Non-founder share of network	5.21%	$\leq 0.30\%$

Holders keep their coins in absolute terms and lose roughly $17\times$ of their relative position. That is the arithmetic consequence of preserving absolute balances while the supply stays at 21 B. It is a legitimate choice — it is what "preserved in absolute terms" means — but it should be published in exactly these terms rather than as "your balance is preserved", which is true and misleading at the same time.

4A. Concentration under the carried-over balance

The carried-over founder balance is **liquid at slot 0**, so the answer to §0.1 changes and it should be stated in numbers rather than characterised.

Circulating at slot 0	5,033,884,800 BLCH (carryover 3.77 B + liquidity 1.05 B + marketing TGE 0.21 B)
Founder liquid at slot 0	3,546,175,400 BLCH
Founder share of circulating	70.4%

Gate G2 requires the largest holder to hold under 25% of active stake. On this schedule:

	Founder share of circulating
Year 1	58.0%
Year 2	41.6%
Year 3	31.5%
Year 5	25.2%
Year 10	20.0%
Year 40	16.9%

G2 is not met until roughly year five, and only if every other bucket unlocks and the validator emission accrues to independent parties as modelled. A draft that cliffed the founder's entire position bought a genesis where the founder held no spendable stake at all; carrying the balance across liquid gives that up.

Two things soften it and should be said alongside the number. The new 17% grant is locked for a decade and vests across forty years — far beyond any market benchmark, and the strictest schedule on the chain. And the §4.1 machinery distinguishes **liquid** from **stakeable**: a carried-over balance can be spendable while remaining ineligible to stake. Keeping the carryover liquid does not by itself decide that it votes. That is a separate decision, still open, and it is the one that determines whether the activation gates are reachable before year five.

5. What this does to the concentration problem

This section argued that the relaunch was the strongest answer to the concentration objection. Two later decisions changed the answer, and both weakened it — so the honest version replaces the argument rather than restating it.

The PoS migration design opens (§0.1) with the objection that ~94% of supply sits at one address, so stake-weighted consensus would hand the chain to the founder. Where V4 now lands:

	Genesis-3 today	V4 at genesis	V4 at year 50
Founder — spendable	94.3%	16.89% (carryover, liquid)	26.89%
Founder — locked	—	10.00% (10-yr cliff, 40-yr vest)	0%
Insiders total	94.3%	25.0% liquid (Foundation) + 16.89%	55.97%
Validators (earned)	—	0%, growing	43.03%

An earlier draft cliffed the founder's whole position and could claim **no spendable founder stake at genesis** — a consensus-enforced answer, and a much stronger one than this. Carrying the balance across liquid gave it up: the founder holds 16.89% of supply, spendable from slot 0, and 70.4% of what is actually circulating then.

What remains true is narrower and still worth something. The relaunch takes the founder from 94.3% of supply to 16.89% spendable, which is a $5.6\times$ reduction in concentration that no amount of coin-marking on the live chain could have achieved. And the new grant carries the strictest schedule on the chain — ten years locked, forty vesting, fully vested at year 50.

6. Emission

- Validator allocation: **9,036,115,200 BLCH** over **40 years**.
- At 30 s slots: 42,076,800 slots in 40 years.
- **Average 215 BLCH per block.**

6.1 Curve — decided: 10% annual disinflation

Founder constraint: **annual inflation under 7%**. Combined with the decentralisation requirement from §7A, that pins the curve almost exactly.

Curve	Year-1 inflation	Decentralisation gate
Flat, 215 BLCH/block	1.08%	Fails — validators never outpace insider unlocks
Halving every 4 years	5.38%	Passes, but revenue halves on scheduled dates
Decay, 8%/year decline	3.56%	Fails at month 36 — too flat
Decay, 10%/year decline	4.37%	Passes
Decay, 12%/year decline	5.19%	Passes, but the shape, not the ceiling, is what rules it out

Adopted: reward declines 10% per year, constant within each year, summing to exactly the 9,036,115,200 allocation across 40 years.

Year	BLCH/block	Inflation (of total supply)
1	871.90	4.37%
5	572.05	2.87%
10	337.79	1.69%
20	117.78	0.59%
40	14.32	0.07%

Truncation residual across the whole 40-year schedule: **889,200 sat (0.0089 BLCH)** — under the allocation, never over.

The denominator is load-bearing. These figures are issuance over **total supply**, which is how Solana and Ethereum report inflation. Measured against *circulating* supply the same curve reads over 100% in year one — not because issuance is high, but because almost every allocation is still vesting at genesis, so the float is only ~6.3 B. Any public figure must say which denominator it uses.

A hard 7%-of-*circulating* rule was modelled and rejected: it emits only 0.44 B in year 1 and 6.65 B by year 5, so validators never outpace the insider unlock schedule, and the concentration gate fails outright.

6.2 Why not a halving

Neither Ethereum nor Solana has one — halving is essentially a Bitcoin convention. Ethereum's issuance is dynamic (it scales with total ETH staked, with base fees burned on top, so net issuance can go negative); Solana runs smooth disinflation, 8% initial declining 15%/year to a 1.5% floor. The curve adopted here is the Solana shape adapted to a hard cap: no floor, no tail, the schedule simply ends.

Beyond convention, a halving is a scheduled date on which every validator's revenue drops by half at once, and marginal operators exit together. Continuous decay has no such edges — the same reasoning that put founder vesting on a per-slot line rather than monthly tranches.

Hard cap — a reversal worth recording. The current design is explicitly *not* hard-capped:

`tokenomics_v2.rs` runs a perpetual 100 BLCH tail (Monero-style), and ADR-035 lowered that floor to 60. V4 replaces it with a fixed cap and a fee-only end state. That is a deliberate and defensible choice, but it inherits the fee-only security-budget question: after year 40, the entire cost of consensus must be covered by fee revenue, and if fees are thin the validator set shrinks to whatever fees sustain. The tail existed to avoid exactly that. The reversal should be recorded in an ADR that retracts the tail rationale rather than silently superseding it.

6.3 Validator revenue — Solana model

Adopted: validator revenue mirrors Solana's, which has three parts.

Stream	Rule	Note
Inflation rewards	Pro-rata to all active stake , scaled by attestation credits earned; validator takes a commission on delegated stake only	Producing blocks is not how you get paid — being staked behind a performing validator is
Base fee	50% burned , 50% to the block producer	Solana's split
Priority fee	100% to the block producer	Solana moved from 50/50 to 100% with SIMD-0096

Commission is uncapped by consensus (Solana allows 0–100%; single digits in practice). A cap is trivially evaded by an operator running its own delegation front-end, so the rule is disclosure rather than limitation: wallets and the explorer must surface the rate prominently.

Yield versus inflation. These are different numbers and both get quoted. At year-1 issuance with two thirds of supply staked — Solana's rough ratio — the nominal staking yield is **6.55%**, against **4.37%** inflation. A staker's real position is what remains after dilution; a non-staker is diluted by the full 4.37%.

For reference, Bloch's year-1 inflation of 4.37% sits just below Solana's current 5.5–5.9%.

6.3.1 Delegation — implemented (`crates/bloch-pos-committee/src/delegation.rs`)

Commission is meaningless without delegated stake, and pro-rata-to-all-stake rewards only make sense if stake can sit behind an operator without running one. **The Solana revenue model cannot be adopted without adding delegation**, which the PoS design does not currently have: validators deposit directly with a 100,000 BLCH minimum.

Four rules make delegation safe to add:

Rule	Value	Why
Warm-up / cool-down rate limit	9% of active stake per epoch	Committees are stake-weighted, so instant activation is instant control. Matches Solana
Delegated stake counts toward the per-validator cap	1% of active stake	Delegation must not be a route around §4.1
Delegators are exposed to slashing	pro-rata	Otherwise delegation is all yield and no risk, and nobody cares who they delegate to
Tainted coins cannot delegate	—	§4.1 follows coins, not accounts; otherwise delegation launders eligibility

Two implementation findings worth recording, because both were wrong in the first cut:

- **The cap must be resolved by fixed-point iteration, not measured against the uncapped total.** Against the uncapped total, an operator holding 90% of raw stake among a hundred is clamped to 9.99M versus 1M for everyone else — still ten times any peer, 9.2% of effective weight, present in over half of all committees. The cap's strength degrades exactly as concentration rises, which is when it is needed. Iterating to a fixed point clamps that same operator to 1.0%, level with a normal validator: a ninefold improvement. The iteration is safe to specify — clamping only lowers the total, which only lowers the cap, so it is monotone and converges — and the round count is fixed so every node stops at the same number.

- **The rate limit needs a liveness escape.** A strict 9% budget deadlocks forever on any single delegation larger than 9% of active stake, which on a young network is most of them, and deadlocks at genesis where active stake is zero. Genesis is unlimited, and thereafter the head of the queue always progresses even if it exceeds the budget — bounding disruption to one record per epoch while guaranteeing the queue drains.

Decentralisation cuts both ways, and this is now measurable rather than asserted:

`Registry::top_share_bps` and `Registry::nakamoto_coefficient` compute gates G2 and G3 directly from the delegation set, closing the gap §7A left open. The Nakamoto coefficient is computed at the **one-third** threshold — the point at which a two-thirds quorum can be stalled — not one half, which would flatter the figure.

The honest limit: those metrics see the *operator* view, which is what consensus sees. They cannot see one beneficial owner standing behind several delegators, and no on-chain metric can.

6.3.2 Fee policy — decided: two eras

Era	Base fee	Priority fee
During emission (40 years)	50% burned, 50% to producer	100% to producer
After emission	100% to producer, no burn	100% to producer

The burn is the deflationary counterweight that gives the hard cap meaning while new supply is still arriving. Once issuance stops, fees are the entire security budget, and burning part of the only remaining revenue would shrink the validator set for no gain. The switch is at exactly the slot emission stops, so there is never a window with both issuance and a burn, nor one with neither.

Consequence for the supply figure — this changes what "100 billion" means. Burned fees are permanently destroyed, so total supply never reaches 21,000,000,000. The correct statement is:

21,000,000,000 BLCH is the **maximum ever issued**. Circulating supply settles at that figure minus everything burned during the 40-year emission era, and is fixed thereafter.

Any published supply number must use "maximum issued", not "total supply" — the two diverge from the first burned fee onward, and the gap only grows.

A cliff worth naming. At the era boundary, validator revenue loses the year-40 emission (85.10 BLCH/block) and gains only the other half of the base fee. Unless fee revenue by year 40 is comparable to 85 BLCH/block, this is a step down in validator income, and it is inherent to any hard cap rather than to this fee policy — the perpetual tail in V2 existed precisely to avoid it. Whether fee revenue reaches that level by year 40 is unknowable now; what is knowable is that the chain should be monitoring the ratio long before it matters.

7. Vesting — schedules and their basis

Schedules follow prevailing market practice for recent L1 launches.

Bucket	Genesis	Cliff	Linear	Total	Market basis
Founder — new grant (10%)	0%	120 mo (10 yr)	480 mo (40 yr)	50 yr	The V2 premine schedule, restored. Far beyond any market benchmark and the strictest on the chain — deliberately, since the carried-over balance arrives liquid and this grant is the part that can still be made to wait
VC / hedge funds	0%	12 mo	24 mo	3 yr	12-month cliff is the standard among recent L1s (Sui Series A and B both cliff at 12 months); investor vests typically run 2–3 years
Team	0%	18 mo	36 mo	4.5 yr	Institutional standard is 12-month cliff + 36-month linear; 18 months is "defensible and increasingly expected" where institutional investors participate, and it keeps the team cliff off the VC cliff month
Marketing	25%	—	24 mo	2 yr	Listing and launch spend is commonly unlocked at TGE for launch momentum; ongoing programmes vest over ~24–25 months
Liquidity	100%	—	—	—	Liquidity is conventionally 100% unlocked at TGE — vesting it defeats its purpose
Holders	100%	—	—	—	Founder decision: carried-over balances are not vested

Cliffs are staggered on purpose. The most cited failure mode in vesting design is the *cliff wall* — several buckets beginning to unlock in the same month, concentrating sell pressure on one date. VC (12), team (18) and founder (24) are six months apart, so unlocks arrive as a stream.

Where this sits against peers. Insider share here (founder + VC + team + marketing = 41%) falls between Aptos (~32.5% team plus investors) and Celestia (~53%). The VC allocation at 10% is below Sui's 14.1% for private investors.

7B. The four Foundation buckets, one by one

VC, team, marketing and liquidity are held by the Foundation (`BLOCH-ENTITY-STRUCTURE.md` §3). Together they are **6,090,000,000 BLCH — 29.00% of supply**, of which **1,260,000,000 is liquid at genesis**.

VC / crypto hedge funds — 2,100,000,000 (10.00%)

Sold to funds; the Foundation is the counterparty of the round. **Nothing is liquid at genesis:** 12-month cliff, then 24 months linear, fully vested at year 3. This is the allocation that makes the Phase-0 legal review blocking rather than precautionary — selling to investors with a return expectation is what moved BLCH toward the centre of the investment-contract test, and it is why ADR-036 retracted the ownerless thesis.

Development team — 2,100,000,000 (10.00%)

Held by the Foundation and granted to individuals; once granted, individuals hold their own. **Nothing liquid at genesis:** 18-month cliff, then 36 months linear, fully vested at year 4.5. The cliff sits 6 months after the VC cliff on purpose — the *cliff wall*, several buckets unlocking in the same month, is the most cited failure mode in vesting design, so VC (12), team (18) and founder (120) never share a month.

Marketing — 840,000,000 (4.00%)

210,000,000 liquid at genesis (25%), for listing fees and launch spend; the remaining 630,000,000 vests linearly over 24 months. The split follows ordinary practice: launch spend is immediate by nature, ongoing programmes are not.

Liquidity — 1,050,000,000 (5.00%)

100% liquid at genesis. Deployed to exchange order books and AMM pools. Vesting a liquidity bucket defeats the purpose of having one, so this is the one allocation where full unlock is not a concession but the function.

Consolidated

Bucket	BLCH	Share	Liquid at genesis	Fully vested
VC	2,100,000,000	10.00%	0	year 3
Team	2,100,000,000	10.00%	0	year 4.5
Marketing	840,000,000	4.00%	210,000,000	year 2
Liquidity	1,050,000,000	5.00%	1,050,000,000	genesis
Total	6,090,000,000	29.00%	1,260,000,000	

The number worth noticing

Circulating supply at slot 0 is 5,033,884,800 BLCH — the carryover plus these 1,260,000,000. So the Foundation's liquid holding is **exactly 25.0% of circulating at genesis**, sitting precisely on the G2 threshold, and the carryover is the other 75.0%.

Two entities therefore account for the entire genesis float, and the concentration gates cannot be met by either of them changing behaviour — only by validator emission and independent stake diluting both. That is the same conclusion §4A reaches from the carryover side, arrived at from the Foundation side.

7A. The unlock model, run against the PoS gates

Modelling the schedules month by month, treating each bucket as a single entity (worst case), against gate **G2 — no entity above 25% of active stake**:

With the flat emission curve:

Month	Circulating (B)	Validators	Largest bucket	Insiders
6	7.7	8.7%	64.8%	22.7%
12	9.1	14.7%	54.7%	27.3%
24	18.7	14.4%	26.8%	57.2%
36	30.0	13.4%	33.3%	68.9%
60	41.1	16.3%	24.3%	70.8%

Month	Circulating (B)	Validators	Largest bucket	Insiders
120	56.3	23.8%	24.1%	66.8%

G2 fails for roughly the first five years, and insiders peak near 71% of circulating supply. The vesting schedule alone does not fix concentration — it reschedules it. The cause is structural: insiders unlock 41 B over ~12 years while validators earn only 1.34 B/year under a flat curve.

With a front-loaded curve (halving every 4 years):

Month	Circulating (B)	Validators	Largest bucket	Insiders
6	10.4	32.3%	48.0%	16.8%
12	14.5	46.3%	34.4%	17.2%
24	29.4	45.7%	17.0%	36.3%
36	46.2	43.7%	21.7%	44.8%
60	64.6	46.8%	15.5%	45.0%
120	86.6	50.4%	15.7%	43.4%

Validators cross 45% of circulating supply inside two years, the largest bucket stays under 25% from month 24 onward, and insiders peak near 45% rather than 71%. The only breaches are months 6 and 12 — and there the largest bucket is **liquidity**, which disperses to traders and exchanges rather than acting as a single entity.

Conclusion: the emission curve, not the vesting schedule, is the lever that decides whether PoS can ever activate. Open decision #2 in §9 is therefore not a free parameter — it is the most consequential number left in V4, and the recommendation is a front-loaded curve.

Halving parameters, if adopted: initial reward **1,075 BLCH/block**, halving every 4 years, 10 halvings across the 40-year window.

8. Engineering hazards

8.1 u64 headroom — must be addressed before any code lands

At 8 decimal places, 21 B BLCH is 2.1×10^{18} satoshis — **11.38% of u64::MAX**, and it fits inside the signed `int64` the Go SDK uses for `Satoshis` (`sdk/go/models.go:16`).

An earlier draft put the supply at 100 B, which is 10^{19} satoshis: **54.21% of u64::MAX**, so the sum of two large balances approached the wrap point, and it **overflowed int64 by 8%**, silently turning Go SDK aggregates negative. Returning to the V2 nominal removed both hazards at no cost — they were created by the supply figure, not by the design.

The arithmetic stays `u128` regardless, because the danger was never the totals: it is the **products**. A balance times a basis-point figure, or epoch issuance times stake in the reward split, exceeds `u64` long before any balance does.

Options:

Option	Effect	Cost
Keep 8 decimals, move all accumulators to <code>u128</code> (recommended)	Divisibility unchanged; overflow impossible in accounting paths	Audit of every summation over balances; <code>u64</code> stays for individual values
Reduce to 6 decimals	Supply becomes 0.54% of <code>u64::MAX</code>	Loses two orders of divisibility; changes every address/amount format

Recommended: keep 8 decimals, `u128` for every accumulator, plus a `const _: () = assert!(...)` pinning the invariant so a future supply change cannot quietly re-enter the danger zone. This is the same discipline already applied in `crates/block-pos-committee/src/sample.rs`, where cumulative stake is `u128` for exactly this reason.

8.2 Genesis allocation outputs

Six allocations must appear in the genesis block as consensus-recognised outputs with their unlock schedules enforced by consensus, not by promise — the same standard §4.1 of the migration design applies to the old premine. A vesting schedule that lives in a spreadsheet is not a vesting schedule.

9. Open decisions

- ~~Vesting for VC, team, marketing, liquidity~~ — **decided**, §7.
- ~~Emission curve~~ — **decided**: 10%/year smooth disinflation (§6.1).
- Snapshot height, announced in advance (§3).
- Confirmation of pro-rata scale-down for the over-cap case (§3).
- Decimal places and the `u128` accumulator audit (§8.1).
- ADR retracting the perpetual-tail rationale (§6).
- ~~Delegation~~ — **implemented**, §6.3.1, with the concentration gates now computed from the delegation set.
- ~~Fee burn versus "100% of fees"~~ — **decided**, §6.3.2.
- ~~Genesis validator set — who operates them~~: **Decided** (§3.3.1): the founder funds and operates the cohort, under a consensus-enforced cap that tapers to one third within a year.
- ~~The VC allocation against the ownerless thesis~~ — **resolved**: the ownerless thesis is retracted and a Solana-style foundation adopted (ADR-036). What remains is execution: rewrite the public copy before any announcement, and treat the Phase 0 legal review as blocking rather than precautionary.